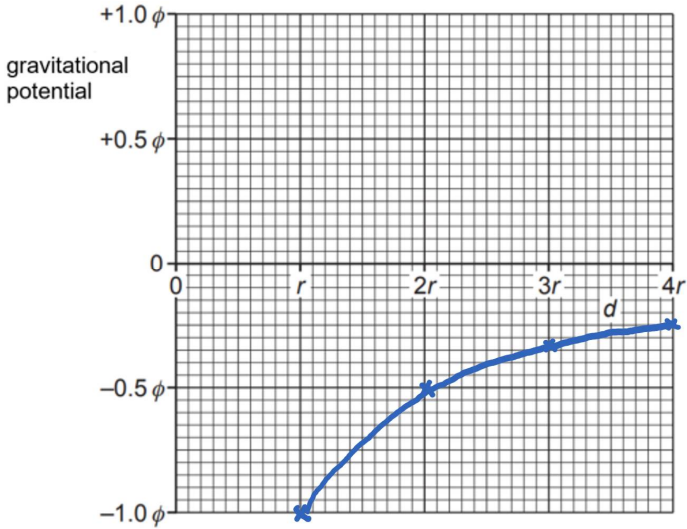


	$= 4/6 \text{ N}$	
2(a)(i)	Work done per unit mass in bringing a small test mass from infinity to that point.	[1] for correct answer
2(a)(ii)	Potential at infinity is taken to be zero. Due to the attractive nature of the gravitational force, work done by an external agent to bring any mass from infinity to that point is always negative. Hence the potential at any point must always be negative.	[1] [1]
2(b)		[1] for correct shape [1] for at least 3 correct plots (i.e. r, $2r$ and $4r$)
2(c)(i)	Gravitational force provide the centripetal force for the spacecraft to move in circular orbit. $\frac{GMm}{R^2} = \frac{mv^2}{R}$ $\therefore v = \sqrt{\frac{GM}{R}}$ $= \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{4(6400000)}}$ $= 3953$ $\approx 4.0 \times 10^3 \text{ m s}^{-1} \text{ (shown)}$	[1] for statement [1] for correct equation and numerical substitution

No.	Solution	Remarks
2(c)(ii) 1.	$\Delta U = U_f - U_i$ $= \left(-\frac{GMm}{R_f} \right) - \left(-\frac{GMm}{R_i} \right)$ $= \left(-\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(8600)}{3(6.4 \times 10^6)} \right) - \left(-\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(8600)}{4(6.4 \times 10^6)} \right)$ $= -4.48 \times 10^{10} \text{ J}$	[1] for correct equation and numerical substitution [1] for correct answer
2(c)(ii) 2.	$v_f = \sqrt{\frac{GM}{R}}$ $= \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{3(6.4 \times 10^6)}}$ $= 4565 \approx 4.57 \times 10^3 \text{ m s}^{-1}$ $\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$ $= \frac{1}{2}(8600)(4565^2 - 3953^2)$ $= 2.24 \times 10^{10} \text{ J}$ Since total energy = kinetic energy + gravitational potential energy and there is more loss in GPE than gain in KE, the total energy decrease.	[1] for correct quantitative method [1] for correct explanation and conclusion